



Machine Learning 1

Lecture 6 - Classification - Probabilistic
Generative Models - Maximum Likelihood -
Discrete Data - Discriminant Functions

Erik Bekkers



Decision theory: Misclassification Rate

$\hat{x}$: Decision boundary

x_0 : Optimal Decision boundary

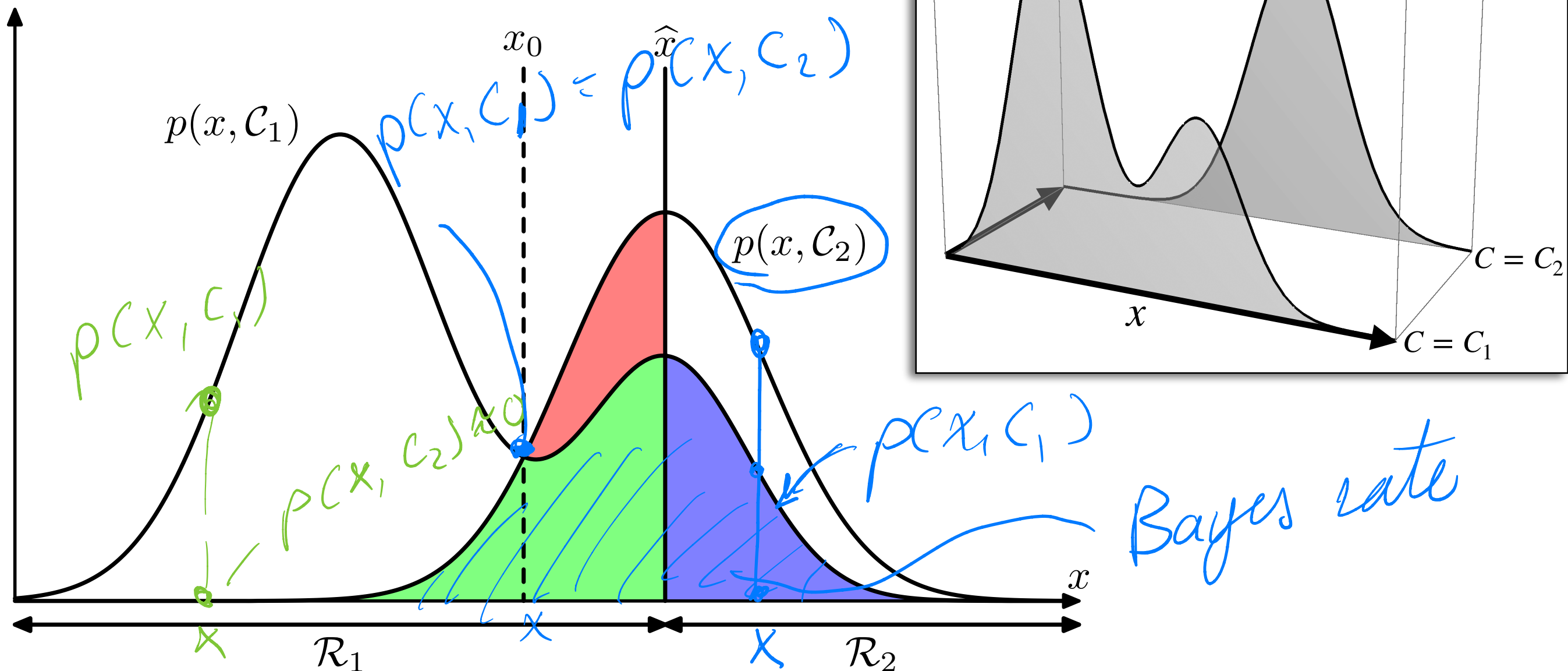


Figure: joint probability distributions and decision boundary (Bishop 1.24)

Classification Strategies

- Discriminant functions

Direct mapping of input to target:

$$t = y(\underline{x}; \underline{w})$$

- Probabilistic discriminative models

Posterior class probabilities:

$$P(C_k | \underline{x}) \stackrel{?}{=} ?$$

- Probabilistic generative models

Class-conditional densities:

$$\left. \begin{array}{l} p(\underline{x} | C_k) \\ p(C_k) \end{array} \right\} \begin{array}{l} \circ p(\underline{x}, C_k) \\ \circ p(C_k | \underline{x}) \end{array}$$

Prior class probabilities:

$$P(C_k | \underline{x}) = \frac{p(\underline{x} | C_k) p(C_k)}{p(\underline{x})}$$

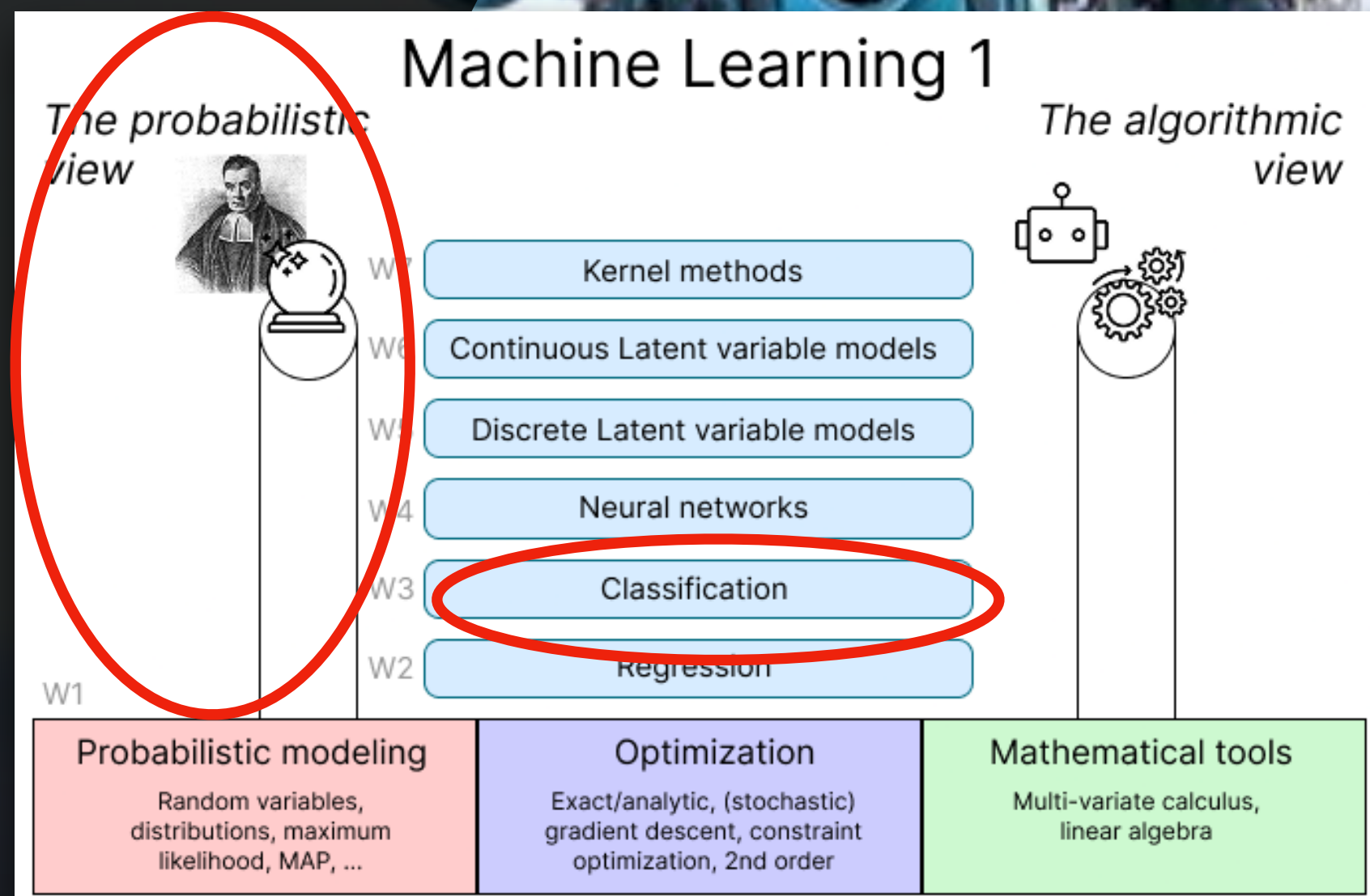


Machine Learning 1

Lecture 5.6 - Supervised Learning
Classification - **Probabilistic Generative Models**

Erik Bekkers

(Bishop 1.5)



Probabilistic Generative Models: general K

- Model:

- Class-conditional densities: $p(\mathbf{x} | C_k)$

- Prior class probabilities: $p(C_k)$

- This gives access to joint and posterior:

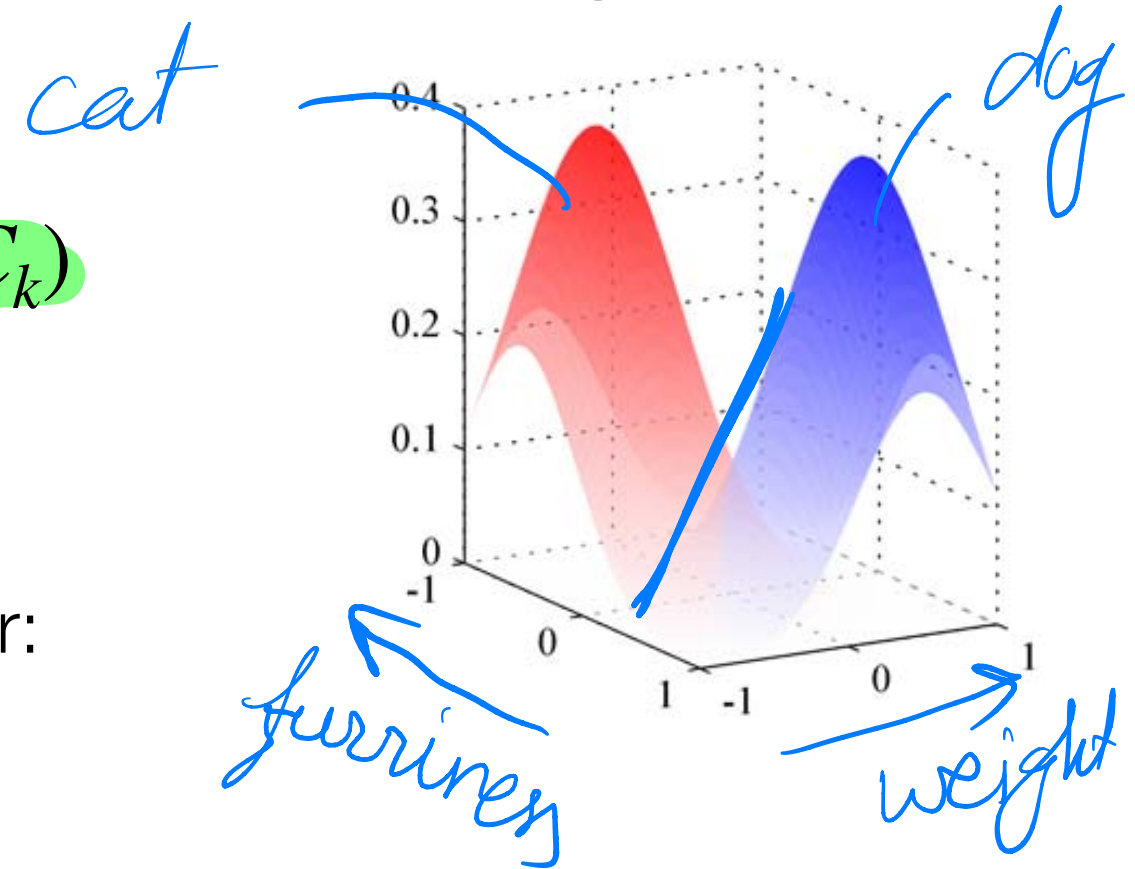
- $p(\mathbf{x}, C_k)$ and $p(C_k | \mathbf{x})$

- Why are we interested in joint or posterior?

- Decision theory tells us that the best prediction for input $\mathbf{x}$, is to choose the class with highest joint $p(\mathbf{x}, C_k) = p(C_k | \mathbf{x})p(\mathbf{x})$

- Or equivalently: choose class with highest posterior $p(C_k | \mathbf{x})$

- Decision boundary between C_k and C_j are at $p(C_k | \mathbf{x}) = p(C_j | \mathbf{x})$



Probabilistic Generative Models: $K = 2$

- ▶ Class-conditional densities: $p(\underline{x} | C_k)$
- ▶ Prior class probabilities: $p(C_k)$
- ▶ Joint distribution: $p(\underline{x}, C_k) = p(\underline{x} | C_k) p(C_k)$
- ▶ Posterior distribution: $K = 2$

Bayes

$$p(C_1 | \mathbf{x}) = \frac{p(\mathbf{x} | C_1)p(C_1)}{p(\mathbf{x} | C_1)p(C_1) + p(\mathbf{x} | C_2)p(C_2)} = \frac{1}{1 + \frac{p(\mathbf{x} | C_2)p(C_2)}{p(\mathbf{x} | C_1)p(C_1)}}$$

$$p(\mathbf{x}) = \sum_k p(\mathbf{x}, C_k)$$

$$= \frac{1}{1 + e^{-a}} =: \sigma(a)$$

$$\text{Log odds } a = \ln \frac{\sigma}{1 - \sigma} = \ln \frac{p(\mathbf{x} | C_1)p(C_1)}{p(\mathbf{x} | C_2)p(C_2)}$$

odds for C_1 vs C_2

$$\frac{p(\underline{x}|C_1)}{p(\underline{x}|C_2)} = \frac{p(\underline{x}|C_1)p(C_1)}{p(\underline{x}|C_2)p(C_2)}$$

What if probs go to 0

Ratio $\rightarrow 0$ or ∞

log-odds

$$\log \frac{p(\underline{x}|C_1)p(C_1)}{p(\underline{x}|C_2)p(C_2)}$$

log odds positive ratio is in favor of C_1

Also note: $-\log \frac{p(\underline{x}|C_1)}{p(\underline{x}|C_2)} = \log \left(\frac{p(\underline{x}|C_2)}{p(\underline{x}|C_1)} \right)$
 $= \log \frac{p(\underline{x}|C_2)}{p(\underline{x}|C_1)}$

Logistic Sigmoid Function

$$\sigma(a) = \frac{1}{1 + \exp(-a)}$$

$$\sigma(-a) = 1 - \sigma(a)$$

$$\sigma'(a) = \sigma(a)(1 - \sigma(a))$$

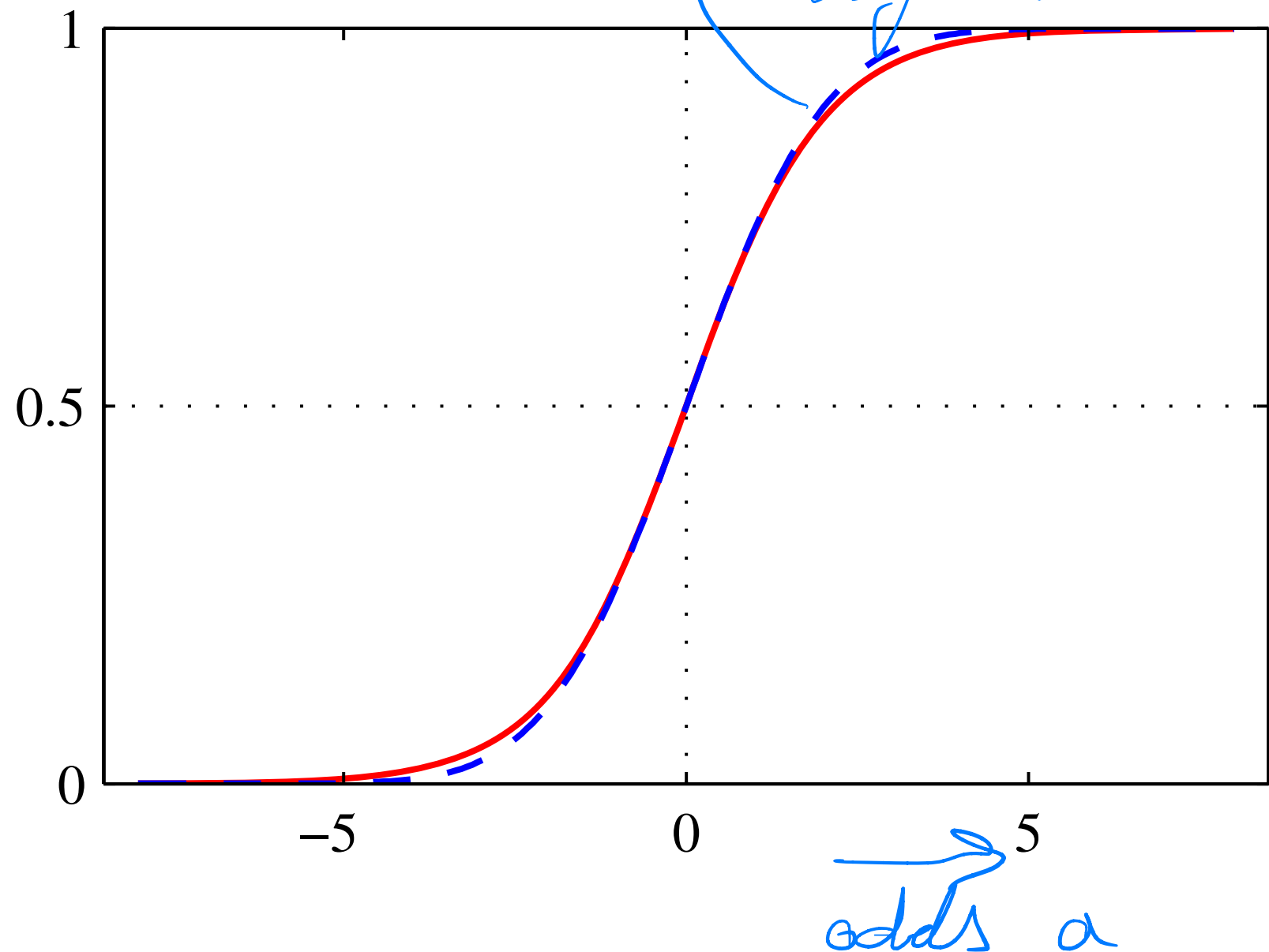


Figure: Logistic Sigmoid function (red) (Bishop 4.9)

Probabilistic Generative Models: general K

- For multiple classes (general K):

$$p(C_k | \mathbf{x}) = \frac{p(\mathbf{x} | C_k)p(C_k)}{\sum_{j=1}^K p(\mathbf{x} | C_j)p(C_j)} = \frac{\exp(a_k)}{\sum_{j=1}^K \exp(a_j)}$$

- Softmax:** if $a_k \gg a_j$ for all $j \neq k$:

$$p(C_k | \mathbf{x}) \approx 1$$
$$p(C_j | \mathbf{x}) \approx 0$$

$$a_k = \ln(p(\mathbf{x} | C_k)p(C_k))$$

called logits

- Note: for $K = 2$:

$$p(C_1 | \mathbf{x}) = \frac{p(\mathbf{x} | C_1)p(C_1)}{p(\mathbf{x} | C_1)p(C_1) + p(\mathbf{x} | C_2)p(C_2)} = \frac{1}{1 + \frac{p(\mathbf{x} | C_2)p(C_2)}{p(\mathbf{x} | C_1)p(C_1)}}$$

$$= \sigma(a), \quad a = a_1 - a_2$$

$$a = \ln \frac{p(\mathbf{x} | C_2)p(C_2)}{p(\mathbf{x} | C_1)p(C_1)}$$

Class Conditional Densities: Continuous Inputs

Modeling Choice

- ▶ Gaussian Class-conditional densities:

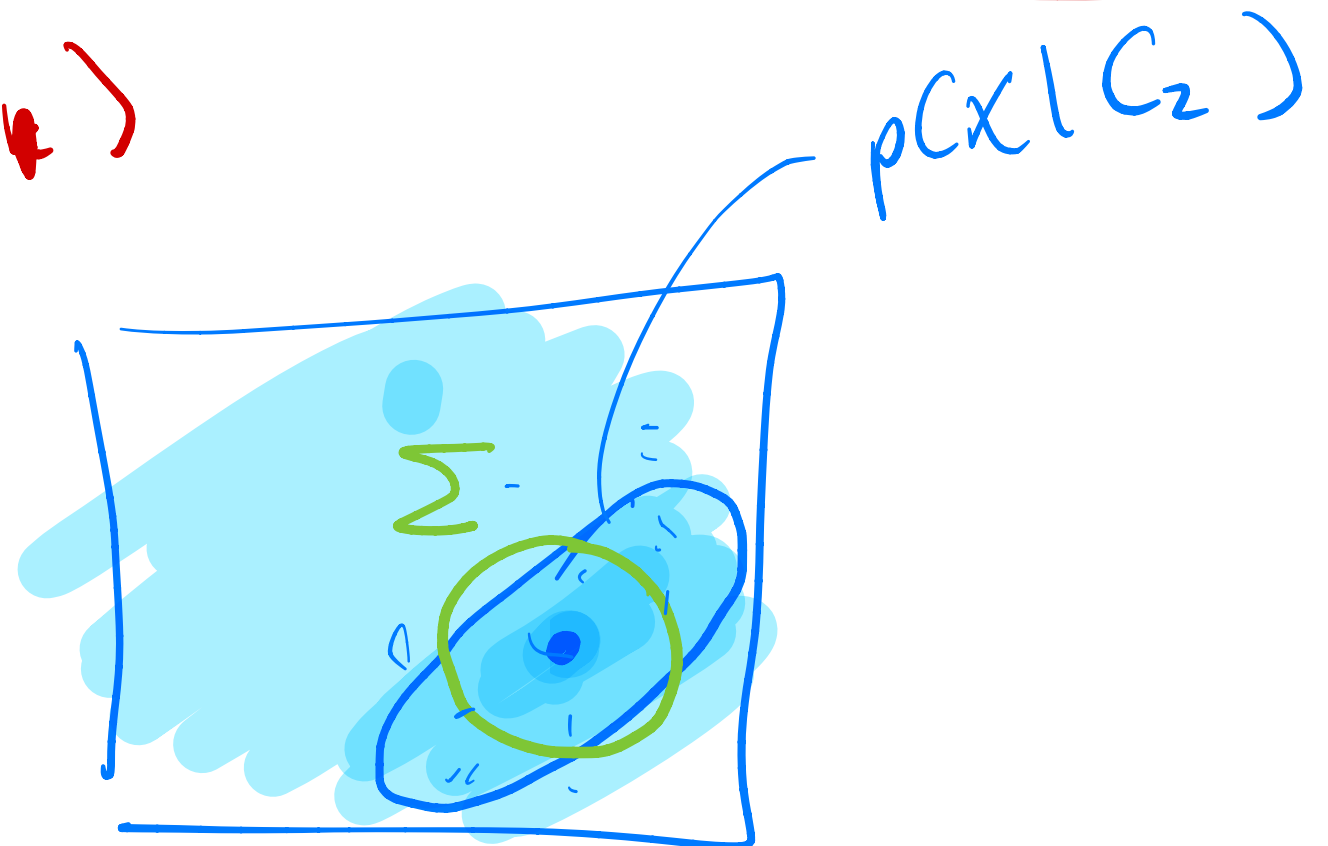
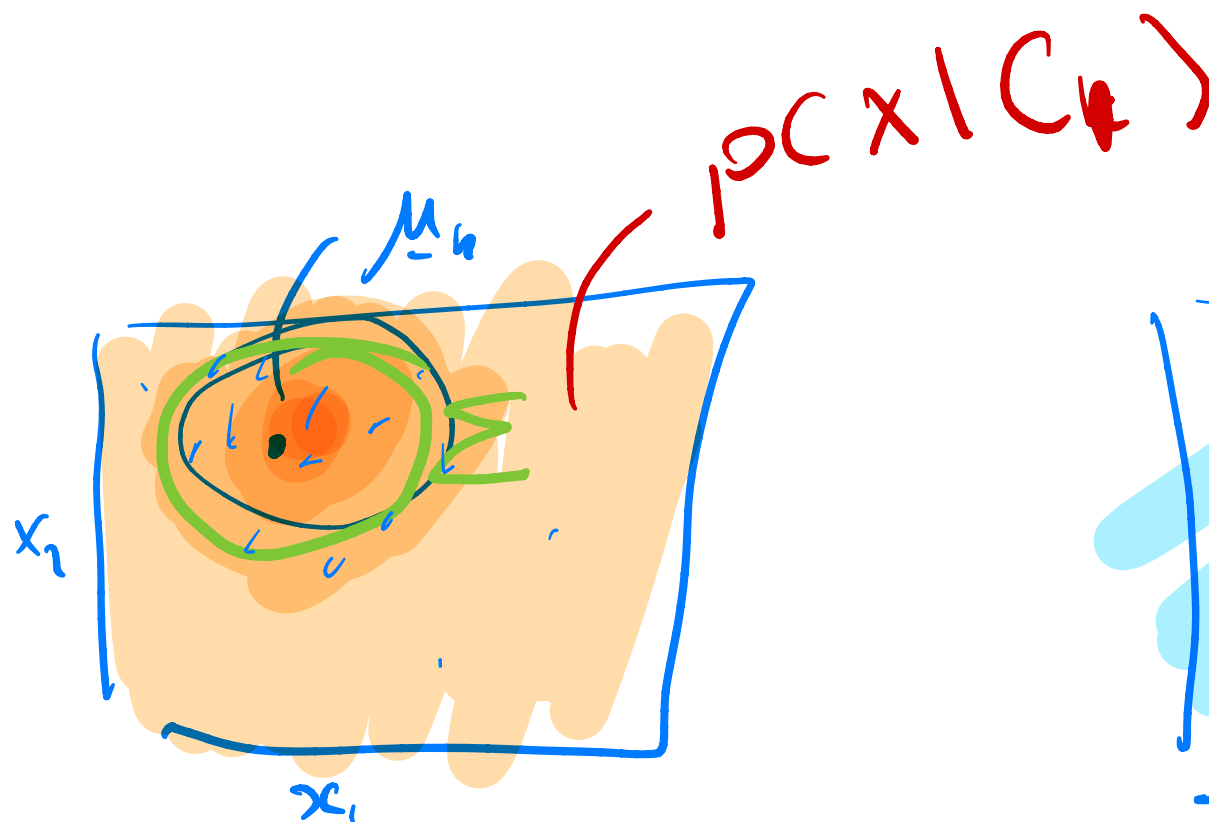
$$p(\mathbf{x} | C_k) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma_k|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \Sigma_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k)\right\}$$

- ▶ Class posteriors simplify with the assumption of shared covariance matrix: $\Sigma_k = \Sigma$

- ▶ For $K = 2$ classes they are given as $p(C_1 | \mathbf{x}) = \frac{1}{1 + \exp(-a)} = \sigma(a)$ with

$$a = \ln \frac{p(\mathbf{x} | C_1)p(C_1)}{p(\mathbf{x} | C_2)p(C_2)}$$

putting bayes rule in logistic form



Class Conditional Densities: Continuous Inputs

- ▶ Gaussian Class-conditional densities:

$$p(\mathbf{x} | C_k) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma_k|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \Sigma_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k)\right\}$$

- ▶ Class posteriors simplify with the assumption of shared covariance matrix: $\Sigma_k = \Sigma$

- ▶ For $K = 2$ classes they are given as $p(C_1 | \mathbf{x}) = \frac{1}{1 + \exp(-a)} = \sigma(a)$ with

$$a = \ln \frac{p(\mathbf{x} | C_1)p(C_1)}{p(\mathbf{x} | C_2)p(C_2)} = \ln \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_1, \Sigma) - \ln \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_2, \Sigma) + \ln \frac{p(C_1)}{p(C_2)}$$

Class Conditional Densities: Continuous Inputs

- ▶ Gaussian Class-conditional densities:

$$p(\mathbf{x} | C_k) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma_k|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \Sigma_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k)\right\}$$

- ▶ Class posteriors simplify with the assumption of shared covariance matrix: $\Sigma_k = \Sigma$

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$$= -\frac{1}{2} \ln |\Sigma| - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_1) + \frac{1}{2} \ln |\Sigma| + \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_2)^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_2) + \ln \frac{p(C_1)}{p(C_2)}$$

$$\cancel{-\frac{1}{2} \mathbf{x}^T \Sigma^{-1} \mathbf{x}}$$

$$+ \mathbf{x}^T \Sigma^{-1} \boldsymbol{\mu}_1$$

$$- \frac{1}{2} \boldsymbol{\mu}_1^T \Sigma^{-1} \boldsymbol{\mu}_1$$

$$\cancel{+\frac{1}{2} \mathbf{x}^T \Sigma^{-1} \mathbf{x}}$$

$$- \mathbf{x}^T \Sigma^{-1} \boldsymbol{\mu}_2$$

$$+ \frac{1}{2} \boldsymbol{\mu}_2^T \Sigma^{-1} \boldsymbol{\mu}_2$$

Class Conditional Densities: Continuous Inputs

- ▶ Gaussian Class-conditional densities:

$$p(\mathbf{x} | C_k) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma_k|^{1/2}} \exp\left\{\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \Sigma_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k)\right\}$$

- ▶ Class posteriors simplify with the assumption of shared covariance matrix: $\Sigma_k = \Sigma$

- ▶ For $K = 2$ classes they are given as $p(C_1 | \mathbf{x}) = \frac{1}{1 + \exp(-a)} = \sigma(a)$ with

$$a = \ln \frac{p(\mathbf{x} | C_1)p(C_1)}{p(\mathbf{x} | C_2)p(C_2)} = \ln \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_1, \Sigma) - \ln \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_2, \Sigma) + \ln \frac{p(C_1)}{p(C_2)}$$

$$= -\frac{1}{2} \ln |\Sigma| - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_1) + \frac{1}{2} \ln |\Sigma| + \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_2)^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_2) + \ln \frac{p(C_1)}{p(C_2)}$$

$$= (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \Sigma^{-1} \mathbf{x} - \frac{1}{2} \boldsymbol{\mu}_1^T \Sigma^{-1} \boldsymbol{\mu}_1 + \frac{1}{2} \boldsymbol{\mu}_2^T \Sigma^{-1} \boldsymbol{\mu}_2 + \ln \frac{p(C_1)}{p(C_2)}$$

$\underbrace{\quad}_{w^T x}$

$+ w_0$

$a(x)$ is a linear model!

Class Conditional Densities: Continuous Inputs

- ▶ Gaussian **Class-conditional densities**:

$$p(\mathbf{x} | C_k) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma_k|^{1/2}} \exp\left\{\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \Sigma_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k)\right\}$$

- ▶ **Class posteriors** simplify with the assumption of shared covariance matrix: $\Sigma_k = \Sigma$

- ▶ For $K = 2$ classes they are given as $p(C_1 | \mathbf{x}) = \frac{1}{1 + \exp(-a)} = \sigma(a)$ with

$$a = \ln \frac{p(\mathbf{x} | C_1)p(C_1)}{p(\mathbf{x} | C_2)p(C_2)} = \ln \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_1, \Sigma) - \ln \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_2, \Sigma) + \ln \frac{p(C_1)}{p(C_2)}$$

$$= -\frac{1}{2} \ln |\Sigma| - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_1) + \frac{1}{2} \ln |\Sigma| + \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_2)^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_2) + \ln \frac{p(C_1)}{p(C_2)}$$

$$= (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \Sigma^{-1} \mathbf{x} - \frac{1}{2} \boldsymbol{\mu}_1^T \Sigma^{-1} \boldsymbol{\mu}_1 + \frac{1}{2} \boldsymbol{\mu}_2^T \Sigma^{-1} \boldsymbol{\mu}_2 + \ln \frac{p(C_1)}{p(C_2)}$$

σ is non-linear

- ▶ **Generalized Linear Model**: $p(C_1 | \mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + w_0)$

$$\mathbf{w} = \Sigma^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$

$$w_0 = -\frac{1}{2} \boldsymbol{\mu}_1^T \Sigma^{-1} \boldsymbol{\mu}_1 + \frac{1}{2} \boldsymbol{\mu}_2^T \Sigma^{-1} \boldsymbol{\mu}_2 + \ln \frac{p(C_1)}{p(C_2)}$$

linear

Example: Linear Discriminant Analysis for K=2

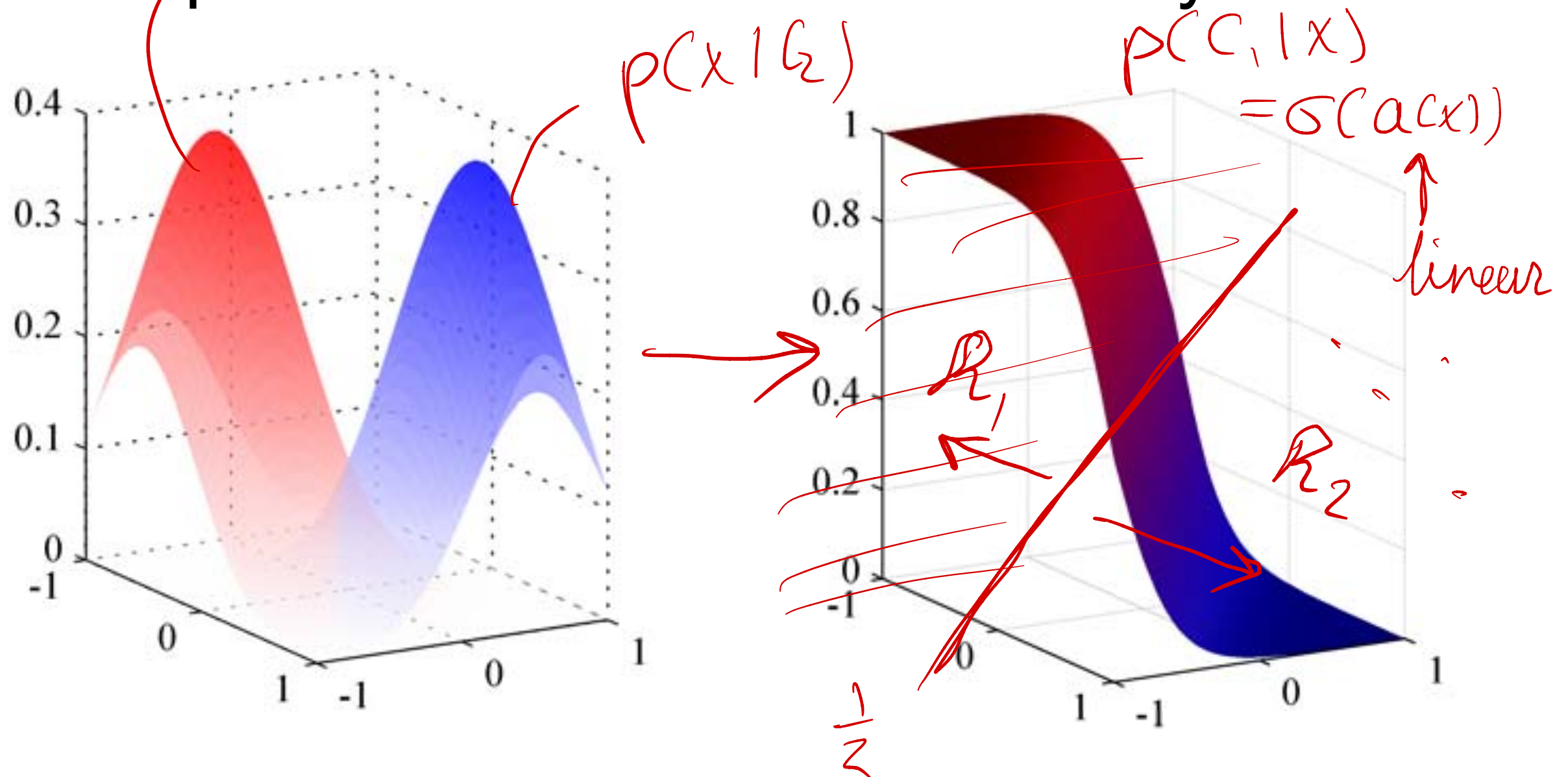


Figure: Left: class conditional densities $p(\mathbf{x} | C_k)$. Right: posterior $p(C_1 | \mathbf{x})$ as sigmoid of linear function of $\mathbf{x}$. (Bishop 4.9)

Linear Discriminant Analysis: General K

- ▶ Gaussian Class-conditional densities & fixed covariance:

$$\Sigma_k = \Sigma$$

$$p(\mathbf{x} | C_k) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma_k|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \Sigma_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k)\right\}$$

- ▶ Posterior distributions:

$$p(C_k | \mathbf{x}) = \frac{\exp(a_k(\mathbf{x}))}{\sum_{j=1}^K \exp(a_j(\mathbf{x}))}$$

one linear model per class

has this form

$$a_k(\mathbf{x}) = \mathbf{w}_k^T \mathbf{x} + w_{k0}$$

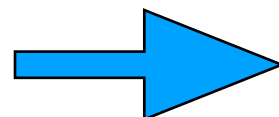
$$\mathbf{w}_k = \Sigma^{-1} \boldsymbol{\mu}_k$$

$$w_{k0} = -\frac{1}{2} \boldsymbol{\mu}_k^T \Sigma^{-1} \boldsymbol{\mu}_k + \ln p(C_k)$$

boundaries!

- ▶ Decision boundary:

$$p(C_k | \mathbf{x}) = p(C_j | \mathbf{x})$$



so linear decision boundaries!

$$a_k(\mathbf{x}) = a_j(\mathbf{x})$$

- ▶ If all covariance matrices are different $\Sigma_k \neq \Sigma_j$ then $a_k(\mathbf{x})$ also contains quadratic terms in $\mathbf{x}$



non-linear d. b.

Example: LDA and QDA

Quadratic
discriminant
analysis

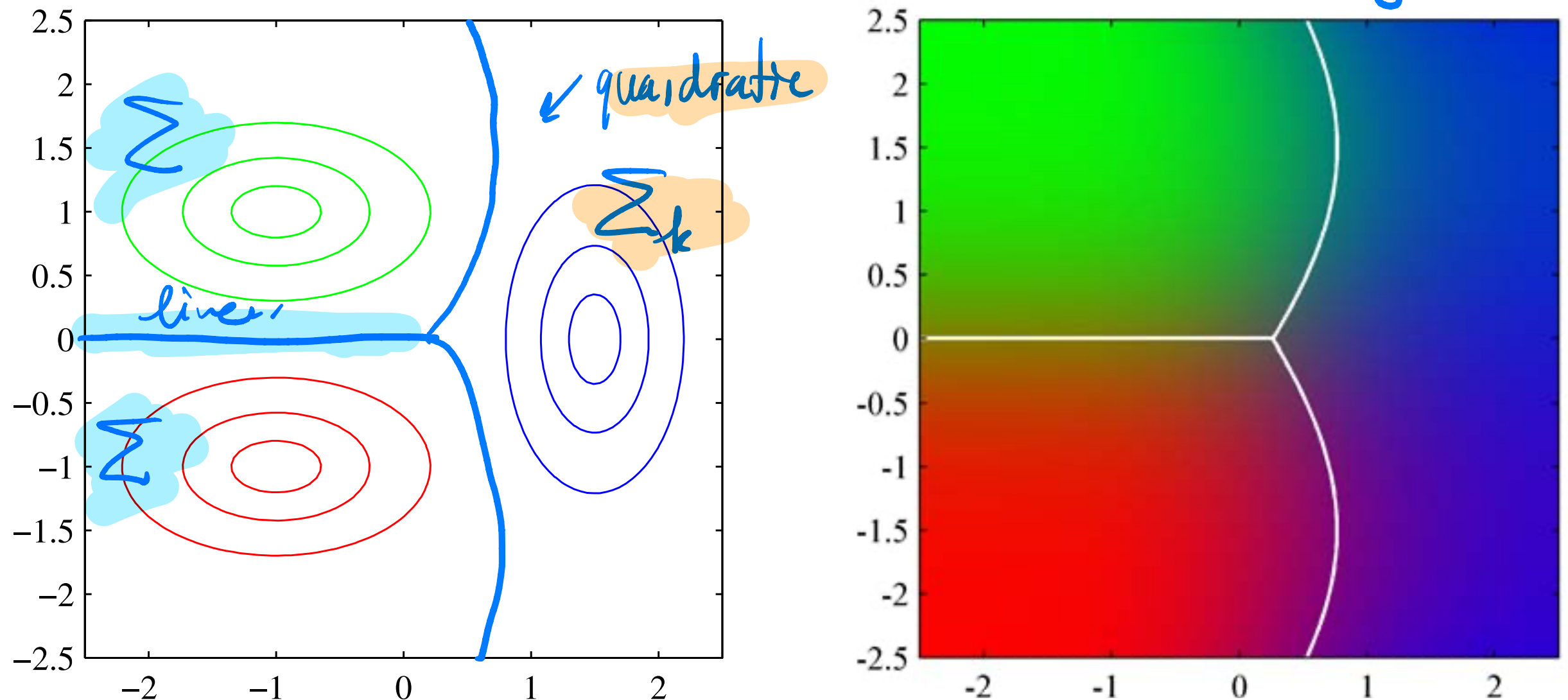


Figure: Left: Gaussian class conditional densities $p(\mathbf{x} | C_k)$, red and green have same covariance matrix. Right: posterior $p(C_k | \mathbf{x})$ distributions (RGB vectors) and decision boundaries. (Bishop 4.9)

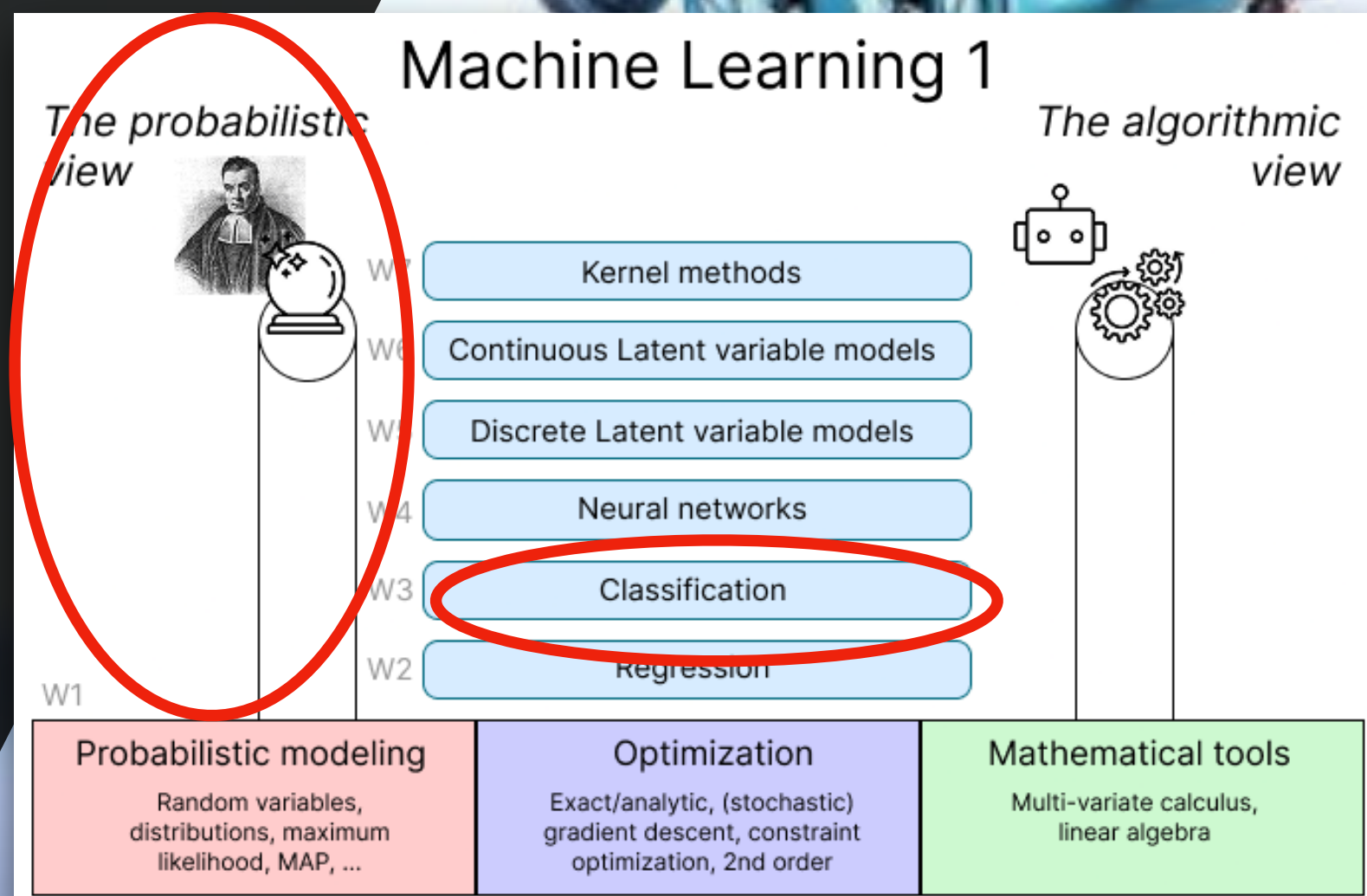


Machine Learning 1

Lecture 6.1 - Supervised Learning
Classification - Probabilistic Generative
Models - Maximum Likelihood Solution

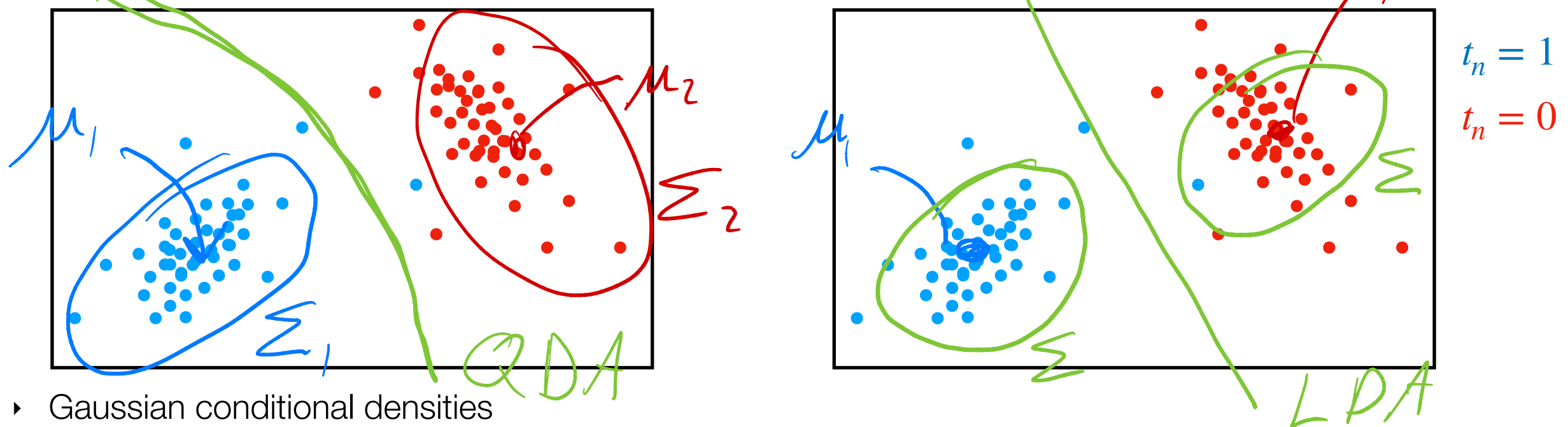
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(Bishop 4.2.2)



LDA: Maximum Likelihood for K=2

- Dataset: input $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_N)^T$, binary targets $\mathbf{t} = (t_1, \dots, t_N)^T$



- Gaussian conditional densities

$$p(\mathbf{x} | C_k) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma_k|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \mu_k)^T \Sigma_k^{-1} (\mathbf{x} - \mu_k)\right\}$$

For now assume model $\Sigma_1 = \Sigma_2 = \Sigma$

- Strategy:

- use maximum likelihood to estimate μ_k , Σ and priors $p(C_k)$
- let q parametrize the prior probabilities with q : $p(C_1) = q$ and $p(C_2) = 1 - q$

one number to parametrize two classes

For $\mathbf{x}_n$ with $t_n = 1$:

$$p(\mathbf{x}_n, C_1) = p(\mathbf{x}_n | C_1)p(C_1) = q N(\mathbf{x}_n | \mu_1, \Sigma)$$

For $\mathbf{x}_n$ with $t_n = 0$:

$$p(\mathbf{x}_n, C_2) = p(\mathbf{x}_n | C_2)p(C_2) = (1 - q) N(\mathbf{x}_n | \mu_2, \Sigma)$$

LDA: Maximum Likelihood for K=2

- Dataset: input $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_N)^T$, binary targets $\mathbf{t} = (t_1, \dots, t_N)^T$

- Gaussian conditional densities

$$p(\mathbf{x} | C_k) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma_k|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \Sigma_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k)\right\}$$

- Strategy:

- use maximum likelihood to estimate $\boldsymbol{\mu}_k$, Σ and priors $p(C_k)$
- let q parametrize the prior probabilities with q : $p(C_1) = q$ and $p(C_2) = 1 - q$

- Likelihood:

$$\begin{aligned}
 p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) &= \prod_{n=1}^N p(\mathbf{x}_n, t_n) = \prod_{n=1}^N p(\mathbf{x}_n | t_n) p(t_n) \\
 &= \prod_{n=1}^N \left[p(\mathbf{x}_n | C_1) p(C_1) \right]^{t_n} \left[p(\mathbf{x}_n | C_2) p(C_2) \right]^{1-t_n} \\
 &= \prod_{n=1}^N \left[q \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma) \right]^{t_n} \left[(1 - q) \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_2, \Sigma) \right]^{1-t_n}
 \end{aligned}$$

Handwritten notes:

- $p(x_1, x_2, x_3, \dots) = p(x_1)p(x_2)p(x_3)\dots$
- $C_1 \Leftrightarrow t=1$
- $C_2 \Leftrightarrow t=0$
- $\{ \text{if } t=1 \text{ then } p(x_n | C_1)p(C_1) \}$
- $\{ \text{if } t=0 \text{ then } p(x_n | C_2)p(C_2) \}$
- $\text{case } t_n=1$
- $\text{case } t_n=0$

LDA: Maximum Likelihood for K=2

$$\log a^t = t \log a$$

► Likelihood

$$p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \prod_{n=1}^N \left[q \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma) \right]^{t_n} \left[(1 - q) \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_2, \Sigma) \right]^{1-t_n}$$

► Log likelihood

$$\ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \sum_{n=1}^N \left[t_n \ln q + t_n \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma) + (1 - t_n) \ln(1 - q) + (1 - t_n) \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_2, \Sigma) \right]$$

Let's estimate q by solving $\frac{\partial}{\partial q} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = 0$

$$q = \frac{N_1}{N_1 + N_2} \quad \frac{\partial}{\partial q} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) =$$

LDA: Maximum Likelihood for K=2

- ▶ Likelihood

$$\frac{\partial}{\partial x} \log x = \frac{1}{x}$$

$$p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \prod_{n=1}^N [q \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma)]^{t_n} [(1 - q) \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_2, \Sigma)]^{1-t_n}$$

- ▶ Log likelihood

$$\begin{aligned} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) &= \sum_{n=1}^N t_n \ln q + t_n \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma) \\ &\quad + (1 - t_n) \ln(1 - q) + (1 - t_n) \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_2, \Sigma) \end{aligned}$$

- ▶ Let's estimate q by solving $\frac{\partial}{\partial q} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = 0$

$$\frac{\partial}{\partial q} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \sum_{n=1}^N \frac{t_n(1-q)}{q(1-q)} \frac{1-t_n}{(1-q)} = \sum_{n=1}^N \frac{t_n(1-q) - (1-t_n)q}{q(1-q)}$$

LDA: Maximum Likelihood for K=2

- Likelihood

$$p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \prod_{n=1}^N [q \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma)]^{t_n} [(1 - q) \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_2, \Sigma)]^{1-t_n}$$

- Log likelihood

$$\begin{aligned} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) &= \sum_{n=1}^N t_n \ln q + t_n \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma) \\ &\quad + (1 - t_n) \ln(1 - q) + (1 - t_n) \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_2, \Sigma) \end{aligned}$$

- Let's estimate q by solving $\frac{\partial}{\partial q} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = 0$

$$\frac{\partial}{\partial q} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \sum_{n=1}^N \frac{t_n}{q} - \frac{1 - t_n}{1 - q} = \sum_{n=1}^N \frac{t_n(1 - q) - (1 - t_n)q}{q(1 - q)} = \sum_{n=1}^N \frac{t_n - q}{q(1 - q)}$$

- Solve for q :

$$\sum_{n=1}^N \frac{t_n - q}{q(1 - q)} = 0$$

LDA: Maximum Likelihood for K=2

- ▶ Likelihood

$$p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \prod_{n=1}^N [q \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma)]^{t_n} [(1 - q) \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_2, \Sigma)]^{1-t_n}$$

- ▶ Log likelihood

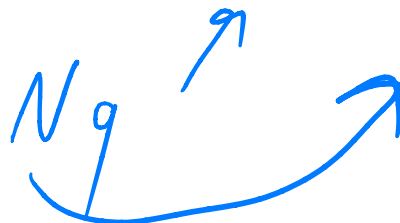
$$\begin{aligned} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) &= \sum_{n=1}^N t_n \ln q + t_n \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma) \\ &\quad + (1 - t_n) \ln(1 - q) + (1 - t_n) \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_2, \Sigma) \end{aligned}$$

- ▶ Let's estimate q by solving $\frac{\partial}{\partial q} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = 0$

$$\frac{\partial}{\partial q} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \sum_{n=1}^N \frac{t_n}{q} - \frac{1 - t_n}{1 - q} = \sum_{n=1}^N \frac{t_n(1 - q) - (1 - t_n)q}{q(1 - q)} = \sum_{n=1}^N \frac{t_n - q}{q(1 - q)}$$

- ▶ Solve for q :

$$\sum_{n=1}^N \frac{t_n - q}{q(1 - q)} = 0 \quad (q \neq 0, q \neq 1) \Rightarrow \sum_{n=1}^N q = \sum_{n=1}^N t_n$$



LDA: Maximum Likelihood for K=2

- ▶ Likelihood

$$p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \prod_{n=1}^N [q \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma)]^{t_n} [(1 - q) \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_2, \Sigma)]^{1-t_n}$$

- ▶ Log likelihood

$$\begin{aligned} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) &= \sum_{n=1}^N t_n \ln q + t_n \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma) \\ &\quad + (1 - t_n) \ln(1 - q) + (1 - t_n) \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_2, \Sigma) \end{aligned}$$

- ▶ Let's estimate q by solving $\frac{\partial}{\partial q} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = 0$

$$\frac{\partial}{\partial q} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \sum_{n=1}^N \frac{t_n}{q} - \frac{1 - t_n}{1 - q} = \sum_{n=1}^N \frac{t_n(1 - q) - (1 - t_n)q}{q(1 - q)} = \sum_{n=1}^N \frac{t_n - q}{q(1 - q)}$$

- ▶ Solve for q :

$$\sum_{n=1}^N \frac{t_n - q}{q(1 - q)} = 0 \quad (q \neq 0, q \neq 1) \Rightarrow \sum_{n=1}^N q = \sum_{n=1}^N t_n \Leftrightarrow q = \frac{1}{N} \sum_{n=1}^N t_n$$

LDA: Maximum Likelihood for K=2

- Log likelihood

$$\ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \sum_{n=1}^N t_n \ln q + t_n \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma) \\ + (1 - t_n) \ln(1 - q) + (1 - t_n) \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_2, \Sigma)$$

- Let's estimate $\boldsymbol{\mu}_1$ by solving $\frac{\partial}{\partial \boldsymbol{\mu}_1} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \mathbf{0}^T$

$$\frac{\partial}{\partial \boldsymbol{\mu}_1} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \frac{\partial}{\partial \boldsymbol{\mu}_1} \sum_{n=1}^N t_n \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma) \\ = -\frac{1}{2} \frac{\partial}{\partial \boldsymbol{\mu}_1} \sum_{n=1}^N t_n (\mathbf{x}_n - \boldsymbol{\mu}_1)^T \Sigma^{-1} (\mathbf{x}_n - \boldsymbol{\mu}_1)$$

LDA: Maximum Likelihood for K=2

- Log likelihood

$$\ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \sum_{n=1}^N t_n \ln q + t_n \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma) \\ + (1 - t_n) \ln(1 - q) + (1 - t_n) \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_2, \Sigma)$$

- Let's estimate $\boldsymbol{\mu}_1$ by solving $\frac{\partial}{\partial \boldsymbol{\mu}_1} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \mathbf{0}^T$

$$\frac{\partial}{\partial \boldsymbol{\mu}_1} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \frac{\partial}{\partial \boldsymbol{\mu}_1} \sum_{n=1}^N t_n \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma) \\ = -\frac{1}{2} \frac{\partial}{\partial \boldsymbol{\mu}_1} \sum_{n=1}^N t_n (\mathbf{x}_n - \boldsymbol{\mu}_1)^T \Sigma^{-1} (\mathbf{x}_n - \boldsymbol{\mu}_1) \quad (\Sigma \text{ is symmetric}) = \sum_{n=1}^N t_n (\mathbf{x}_n - \boldsymbol{\mu}_1)^T \Sigma^{-1}$$

- Solve for $\boldsymbol{\mu}_1$:

$$\sum_{n=1}^N t_n (\mathbf{x}_n - \boldsymbol{\mu}_1)^T \Sigma^{-1} = \mathbf{0}^T \quad (\Sigma \text{ is pos. def.}) \Rightarrow \sum_{n=1}^N t_n \boldsymbol{\mu}_1 = \sum_{n=1}^N t_n \mathbf{x}_n$$

$\mu_1 \cdot \sum_{n=1}^N t_n = N_1$

LDA: Maximum Likelihood for K=2

- Log likelihood

$$\ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \sum_{n=1}^N t_n \ln q + t_n \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma) \\ + (1 - t_n) \ln(1 - q) + (1 - t_n) \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_2, \Sigma)$$

- Let's estimate $\boldsymbol{\mu}_1$ by solving $\frac{\partial}{\partial \boldsymbol{\mu}_1} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \mathbf{0}^T$

$$\frac{\partial}{\partial \boldsymbol{\mu}_1} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \frac{\partial}{\partial \boldsymbol{\mu}_1} \sum_{n=1}^N t_n \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma) \\ = -\frac{1}{2} \frac{\partial}{\partial \boldsymbol{\mu}_1} \sum_{n=1}^N t_n (\mathbf{x}_n - \boldsymbol{\mu}_1)^T \Sigma^{-1} (\mathbf{x}_n - \boldsymbol{\mu}_1) \quad (\Sigma \text{ is symmetric}) \quad \sum_{n=1}^N t_n (\mathbf{x}_n - \boldsymbol{\mu}_1)^T \Sigma^{-1}$$

- Solve for $\boldsymbol{\mu}_1$:

$$\sum_{n=1}^N t_n (\mathbf{x}_n - \boldsymbol{\mu}_1)^T \Sigma^{-1} = \mathbf{0}^T \quad (\Sigma \text{ is pos. def.}) \Rightarrow$$

$$\sum_{n=1}^N t_n \boldsymbol{\mu}_1 = \sum_{n=1}^N t_n \mathbf{x}_n \Leftrightarrow N_1 \boldsymbol{\mu}_1 = \sum_{n=1}^N t_n \mathbf{x}_n$$

$$\boxed{\boldsymbol{\mu}_{1, ML} = \frac{1}{N_1} \sum_{n=1}^N t_n \mathbf{x}_n}$$

average value for $\mathbf{x}_n$ for the cases $t_n=1$

$$\boxed{\boldsymbol{\mu}_{2, ML} = \frac{1}{N_2} \sum_{n=1}^N (1 - t_n) \mathbf{x}_n}$$

LDA: Maximum Likelihood for K=2

- Log likelihood

$$\ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \sum_{n=1}^N t_n \ln q + t_n \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_1, \Sigma) \\ + (1 - t_n) \ln(1 - q) + (1 - t_n) \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_2, \Sigma)$$

- Let's estimate Σ by solving $\frac{\partial}{\partial \Sigma} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \mathbf{O}$

$$\frac{\partial}{\partial \Sigma} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \frac{\partial}{\partial \Sigma} \left[-\frac{N}{2} \ln |\Sigma| - \frac{1}{2} \sum_{n=1}^N t_n (\mathbf{x}_n - \boldsymbol{\mu}_1)^T \Sigma^{-1} (\mathbf{x}_n - \boldsymbol{\mu}_1) \right. \\ \left. - \sum_{n=1}^N (1 - t_n) (\mathbf{x}_n - \boldsymbol{\mu}_2)^T \Sigma^{-1} (\mathbf{x}_n - \boldsymbol{\mu}_2) \right]$$

- Solve $\frac{\partial}{\partial \Sigma} \ln p(\mathbf{t}, \mathbf{X} | q, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \Sigma) = \mathbf{O}$ for $\boldsymbol{\mu}_1$ (Bishop 4.2.2): *+ Notes on Canvas*

$$\Sigma_{\text{ML}} = \frac{N_1}{N} \left[\frac{1}{N_1} \sum_{n=1}^N t_n (\mathbf{x}_n - \boldsymbol{\mu}_{1,\text{ML}})(\mathbf{x}_n - \boldsymbol{\mu}_{1,\text{ML}})^T \right] + \frac{N_2}{N} \left[\frac{1}{N_2} \sum_{n=1}^N (1 - t_n) (\mathbf{x}_n - \boldsymbol{\mu}_{2,\text{ML}})(\mathbf{x}_n - \boldsymbol{\mu}_{2,\text{ML}})^T \right]$$

sample cov of class 1

... class 2

LDA: Maximum Likelihood for K=2

The ML solutions:

$$\underline{\mu_{1,ML}} = \frac{1}{N_1} \sum_{n=1}^N t_n \mathbf{x}_n$$

$$\underline{\mu_{2,ML}} = \frac{1}{N_2} \sum_{n=1}^N (1 - t_n) \mathbf{x}_n$$

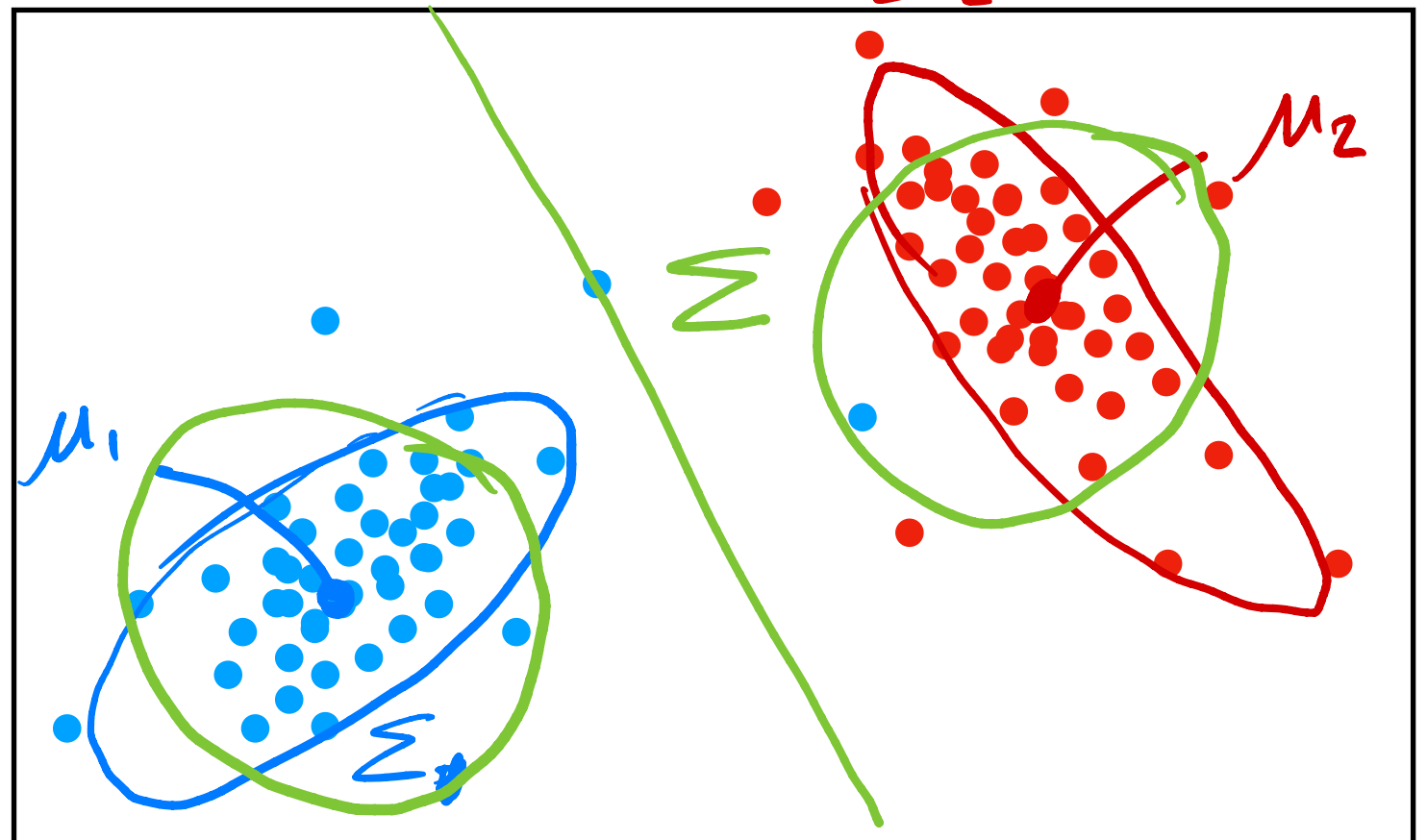
$$q_{ML} = \frac{1}{N} \sum_{n=1}^N t_n = \frac{N_1}{N}$$

$$\underline{\Sigma_{ML}} = \frac{N_1}{N} \left[\frac{1}{N_1} \sum_{n=1}^N t^n (\mathbf{x}_n - \underline{\mu_{1,ML}}) (\mathbf{x}_n - \underline{\mu_{1,ML}})^T \right] + \frac{N_2}{N} \left[\frac{1}{N_2} \sum_{n=1}^N (1 - t^n) (\mathbf{x}_n - \underline{\mu_{2,ML}}) (\mathbf{x}_n - \underline{\mu_{2,ML}})^T \right]$$

For the joint probabilities:

$$\begin{aligned} p(\mathbf{x}, C_1) &= p(\mathbf{x} | C_1) p(C_1) \\ &= \mathcal{N}(\mathbf{x} | \underline{\mu_{1,ML}}, \underline{\Sigma}) q_{ML} \end{aligned}$$

$$\begin{aligned} p(\mathbf{x}, C_2) &= p(\mathbf{x} | C_2) p(C_2) \\ &= \mathcal{N}(\mathbf{x} | \underline{\mu_{2,ML}}, \underline{\Sigma}) (1 - q_{ML}) \end{aligned}$$



LDA: prediction for K=2

- ▶ For new datapoint $\mathbf{x}'$:

derived before

Evaluate: $p(C_1 | \mathbf{x}') = \sigma(\mathbf{w}_{\text{ML}}^T \mathbf{x}' + w_{0,\text{ML}})$

lin model given by ML

$$\text{with } \mathbf{w}_{\text{ML}} = \Sigma_{\text{ML}}^{-1}(\boldsymbol{\mu}_{1,\text{ML}} - \boldsymbol{\mu}_{2,\text{ML}})$$

$$w_{0,\text{ML}} = -\frac{1}{2}\boldsymbol{\mu}_{1,\text{ML}}^T \Sigma_{\text{ML}}^{-1} \boldsymbol{\mu}_{1,\text{ML}} + \frac{1}{2}\boldsymbol{\mu}_{2,\text{ML}}^T \Sigma_{\text{ML}}^{-1} \boldsymbol{\mu}_{2,\text{ML}} + \ln \frac{q_{\text{ML}}}{1 - q_{\text{ML}}}$$

- ▶ Assign $\mathbf{x}'$ to C_1 if $p(C_1 | \mathbf{x}') > \frac{1}{2}$

- ▶ Disadvantage of LDA:

- ▶ Gaussian distribution is sensitive to outliers

(see averaging)

- ▶ Linearity/handcrafted features restrict application

X or use $\phi(K)$

- ▶ Maximum likelihood is prone to overfitting

no regularization



Machine Learning 1

Lecture 6.2 - Supervised Learning

Classification - Probabilistic Generative Models - Discrete Variables (Naive Bayes)

Erik Bekkers

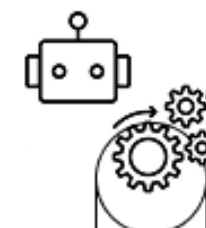
(Bishop 4.2.3)

Machine Learning 1

The probabilistic view



The algorithmic view



Kernel methods

Continuous Latent variable models

Discrete Latent variable models

Neural networks

Classification

Regression

Probabilistic modeling

Random variables,
distributions, maximum
likelihood, MAP, ...

Optimization

Exact/analytic, (stochastic)
gradient descent, constraint
optimization, 2nd order

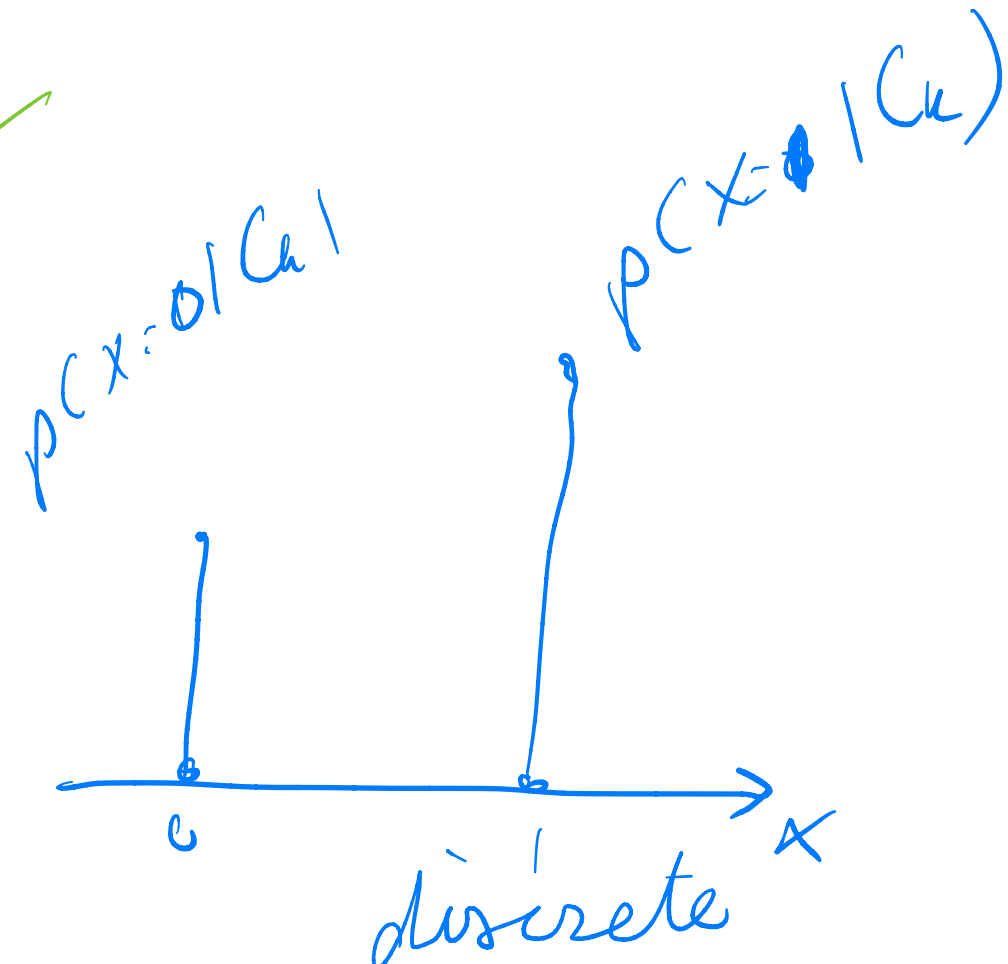
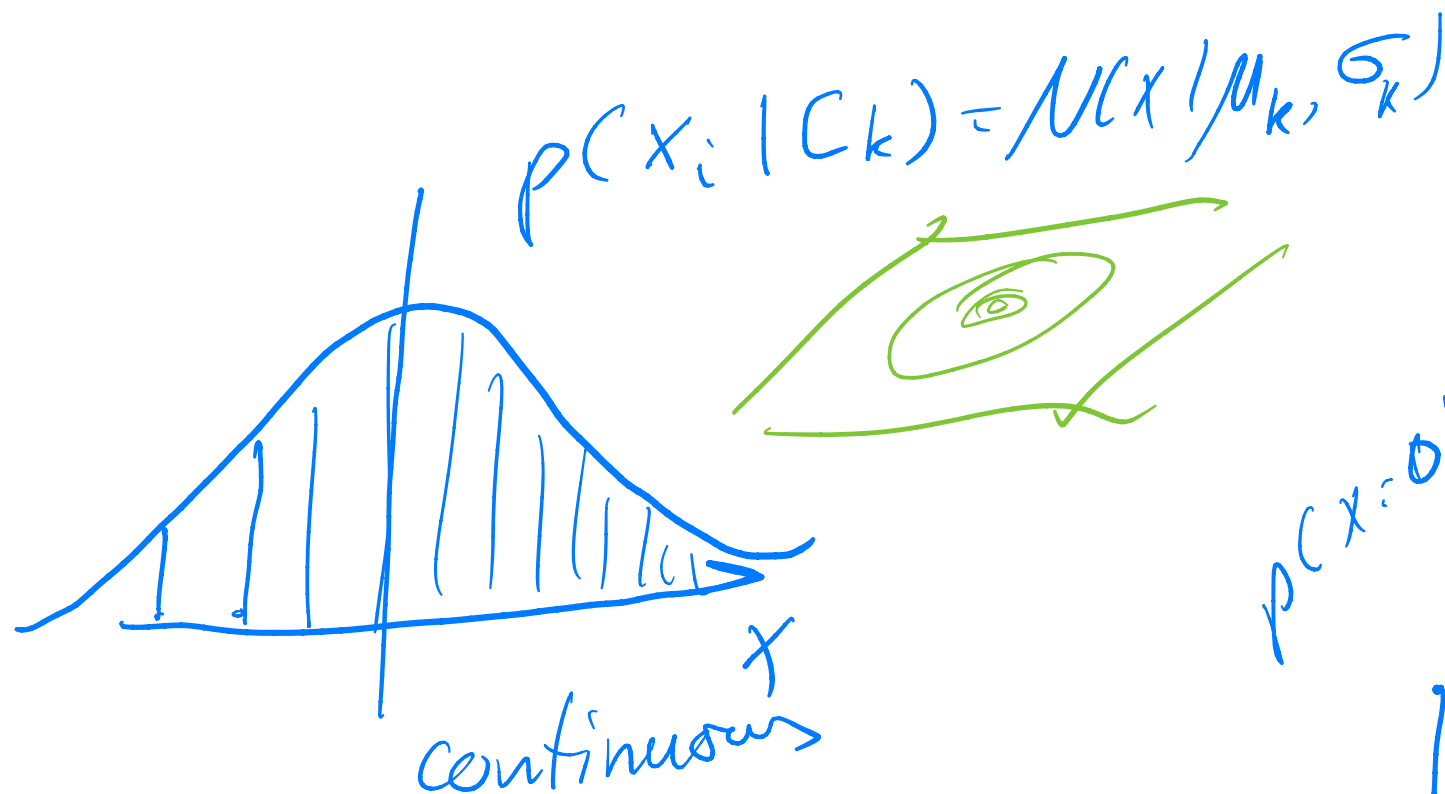
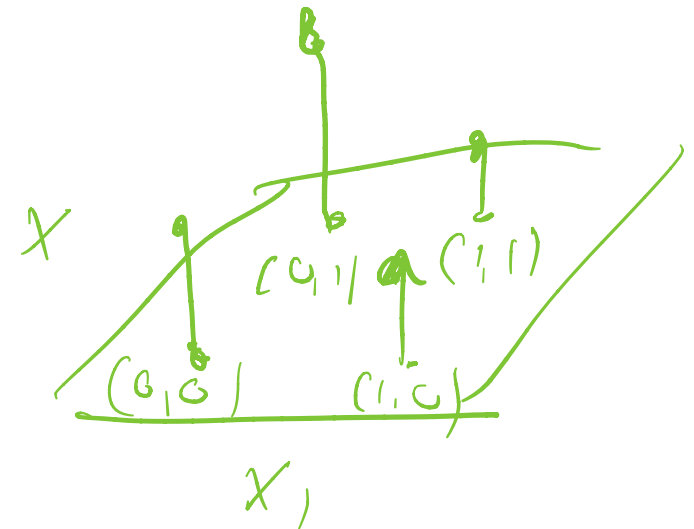
Mathematical tools

Multi-variate calculus,
linear algebra

Probabilistic Generative Models: Discrete ($K = 2$)

- Input: discrete feature vectors $\mathbf{x}_n = (x_1, \dots, x_D)^T = (0, 1, 1, 0, 0, \dots)^T$
 - $x_i \in \{0, 1\}$
- For D-dimensional input # of parameters for $p(\mathbf{x} | C_k)$ per class:

$$2^D - 1$$



Probabilistic Generative Models: Discrete ($K = 2$)

- Input: discrete feature vectors $\mathbf{x}_n = (x_1, \dots, x_D)^T$
 - $x_i \in \{0, 1\}$
- For D-dimensional input # of parameters for $p(\mathbf{x} | C_k)$ per class: $2^D - 1$
- Naive Bayes assumption: feature values are treated as independent when conditioned on class C_k

$$p(\mathbf{x} | C_k) = p(x_1, x_2, x_3, \dots | C_k) = p(x_1 | C_k) p(x_2 | C_k) \dots$$
$$= \prod_i p(x_i | C_k) = \prod_i (\pi_{ki})^{x_i} (1 - \pi_{ki})^{1-x_i}$$

with class conditional parameters π_{ki} .

D

Bernoulli dist.

$$p(x_i = 1 | C_k) = \pi_{ki}$$
$$p(x_i = 0 | C_k) = 1 - \pi_{ki}$$
$$p(x_i | C_k) = \pi_{ki}^{x_i} (1 - \pi_{ki})^{1-x_i}$$

Probabilistic Generative Models: Discrete ($K = 2$)

- Input: discrete feature vectors $\mathbf{x}_n = (x_1, \dots, x_D)^T$
 - $x_i \in \{0, 1\}$
- For D-dimensional input # of parameters for $p(\mathbf{x} | C_k)$ per class: $2^D - 1$
- Naive Bayes assumption: feature values are treated as independent when conditioned on class C_k

$$p(\mathbf{x} | C_k) = \prod_{i=1}^D p(x_i | C_k) = \prod_{i=1}^D \pi_{ki}^{x_i} (1 - \pi_{ki})^{1-x_i}$$

with class conditional parameters π_{ki} .

- Class posterior: $p(C_k | \mathbf{x}) = \frac{\exp(a_k(\mathbf{x}))}{\sum_{j=1}^K \exp(a_j(\mathbf{x}))}$ and prior model parameters $p(C_k) = q_k$
 - $a_k = \ln p(\mathbf{x} | C_k) p(C_k) = \ln p(\mathbf{x} | C_k) + \ln p(C_k)$

$$= \sum_{i=1}^D x_i \ln \pi_{ki} + (1 - x_i) \ln (1 - \pi_{ki}) + \ln q_k$$

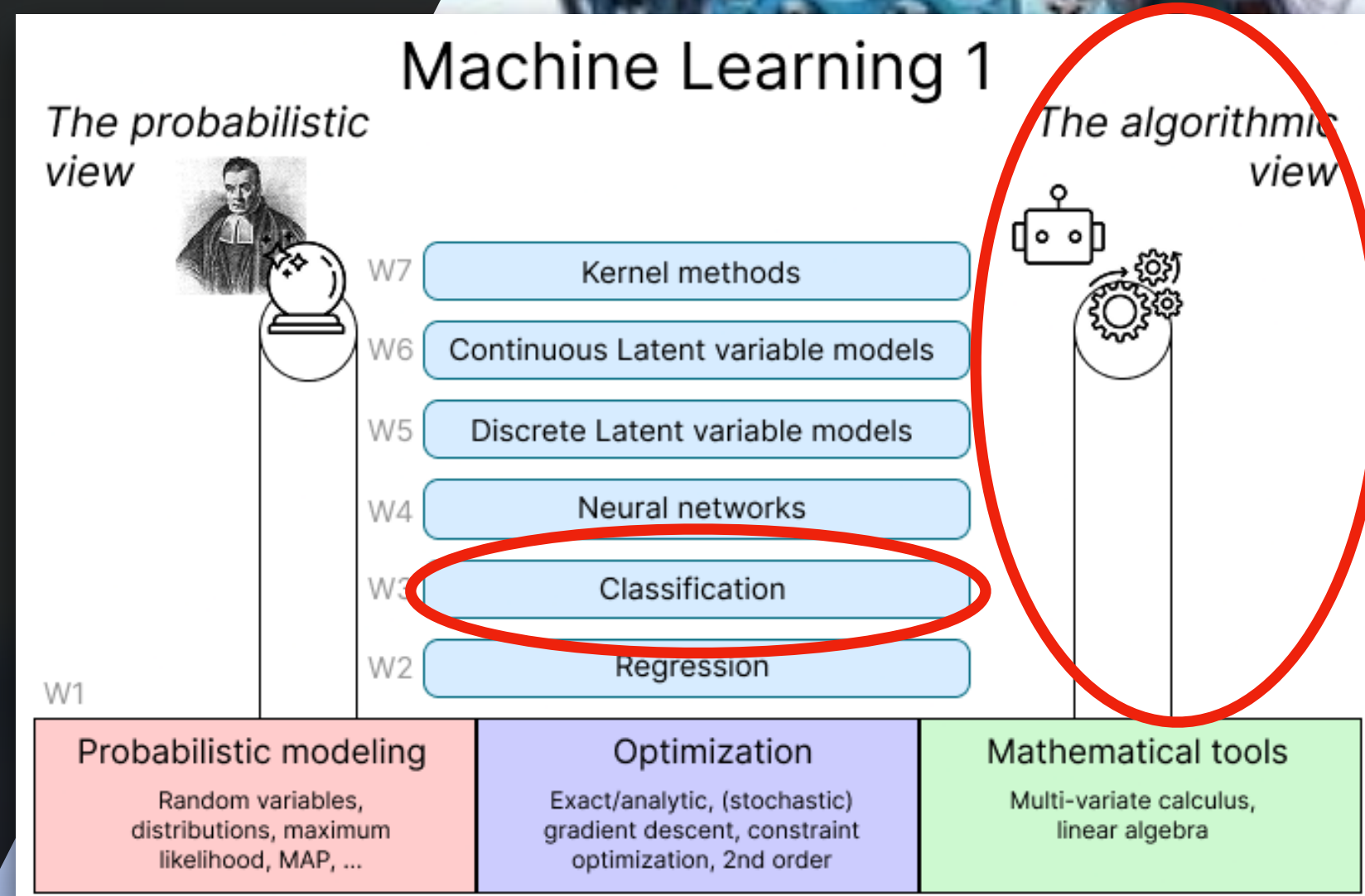


Machine Learning 1

Lecture 6.3 - Supervised Learning
Classification - **Discriminative Models**

Erik Bekkers

(Bishop 4.0, 4.1.1, 4.1.2)



Classification Strategies

- Discriminant functions

Direct mapping of input to target:

$$t = y(\mathbf{x}, \mathbf{w})$$

- Probabilistic discriminative models

Posterior class probabilities:

$$p(C_k | \mathbf{x})$$

- Probabilistic generative models

Class-conditional densities:

$$p(\mathbf{x} | C_k)$$

*means we can
sample from the
learned data
distribution*

Prior class probabilities:

$$p(C_k)$$

LDA

Discriminant Functions: Two Classes

- ▶ Input: $\mathbf{x} \in \mathbb{R}^D$
- ▶ Targets: $t \in \{C_1, C_2\} = t=0, t=1$

Discriminant functions:

- ▶ Direct mapping of input to target

We now consider generalized Linear Models (GLM)

- ▶ $y(\mathbf{x}, \tilde{\mathbf{w}}) = f(\tilde{\mathbf{w}}^T \boldsymbol{\phi}(\mathbf{x}))$
- ▶ with $\boldsymbol{\phi}(\mathbf{x}) = (1, \phi_1(\mathbf{x}), \dots, \phi_{M-1}(\mathbf{x}))^T$
- ▶ Decision boundary: $y(\mathbf{x}, \tilde{\mathbf{w}}) = \text{const}$

"so, why not directly model such a y ?"

Discriminant Functions: Two Classes

- Simplest discriminant function ($f(\mathbf{x}) = x$, canonical basis $\phi_0(\mathbf{x}) = 1$, $\phi_j(\mathbf{x}) = x_j$, $j = 1, \dots, D$)

$$y(\mathbf{x}, \tilde{\mathbf{w}}) = \mathbf{w}^T \mathbf{x} + w_0$$

- Decision boundary: $y(\mathbf{x}, \tilde{\mathbf{w}}) = 0$

$$\underline{\mathbf{w}}^T \underline{\mathbf{x}} + w_0 = 0$$

- Consider 2 datapoints $\mathbf{x}_A$ and $\mathbf{x}_B$ on decision boundary

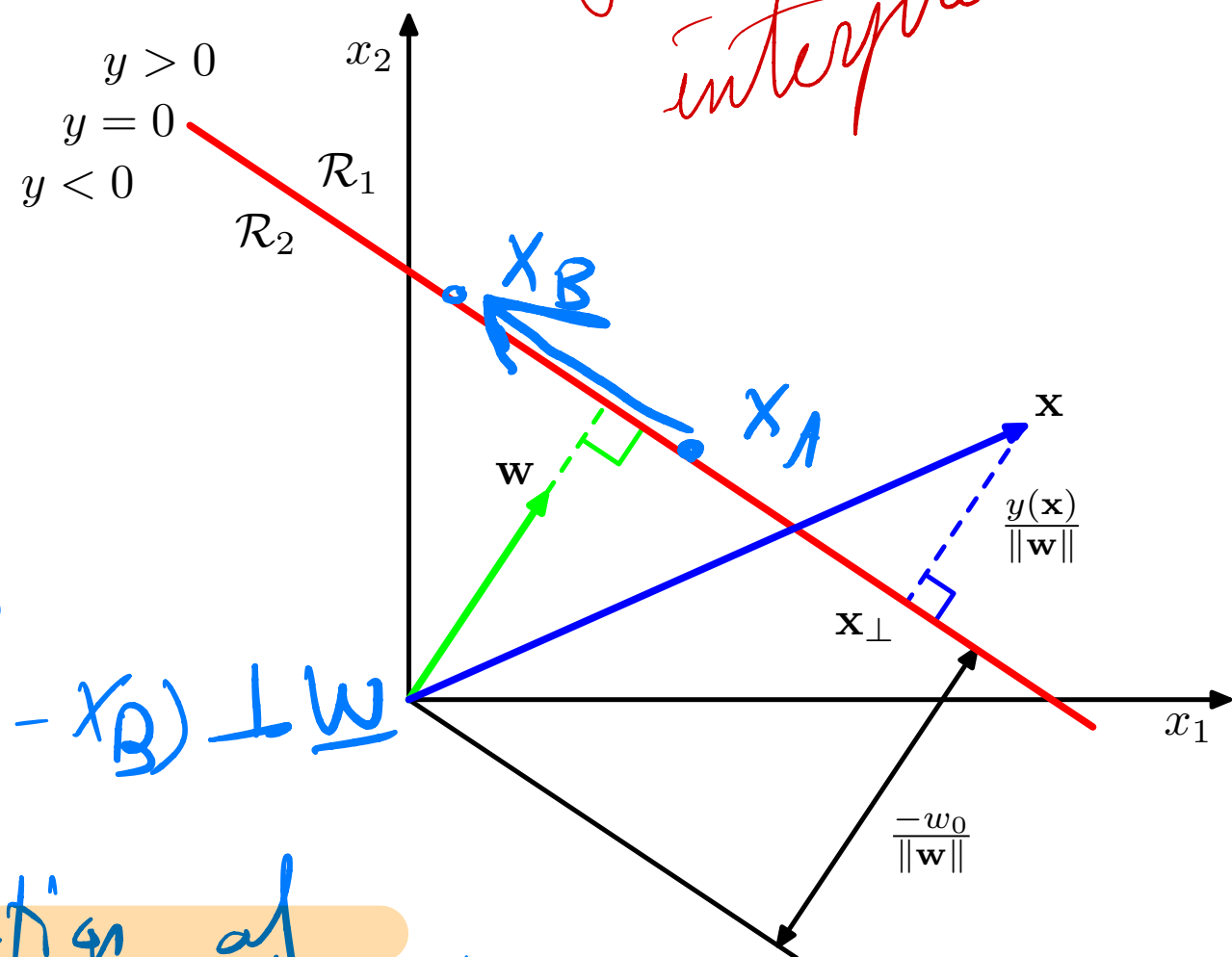
$$\underline{\mathbf{w}}^T \mathbf{x}_A + w_0 = \underline{\mathbf{w}}^T \mathbf{x}_B + w_0$$

$$y(\mathbf{x}_A) = y(\mathbf{x}_B) = 0$$

$$\underline{\mathbf{w}}^T (\mathbf{x}_B - \mathbf{x}_A) = 0$$

i.e.,
 $(\mathbf{x}_A - \mathbf{x}_B) \perp \underline{\mathbf{w}}$

Still usefull to establish some intuition.
 Geometric interpretation



$\underline{\mathbf{w}}$ determines orientation of decision boundary
 "normal vector of the hyperplane"

Figure: 2-class decision boundary (Bishop 4.1) 5

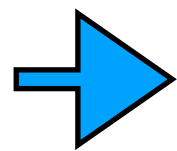
Discriminant Functions: Two Classes

- Take $\mathbf{x}'$ a point on a decision surface: $y(\mathbf{x}') = 0$

$$\mathbf{w}^T \mathbf{x}' + w_0 = 0 \Leftrightarrow \mathbf{w}^T \mathbf{x}' = -w_0$$

- Normal (signed) distance d from origin to decision surface

$$d = \frac{\mathbf{w}^T \mathbf{x}'}{\|\mathbf{w}\|} = \frac{-w_0}{\|\mathbf{w}\|}$$



w shifts the d.b. away from the origin

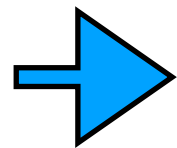
- Normal (signed) distances r from general $\mathbf{x}$ to decision surface:

Let $\mathbf{x} = \mathbf{x}_\perp + r \frac{\mathbf{w}}{\|\mathbf{w}\|}$

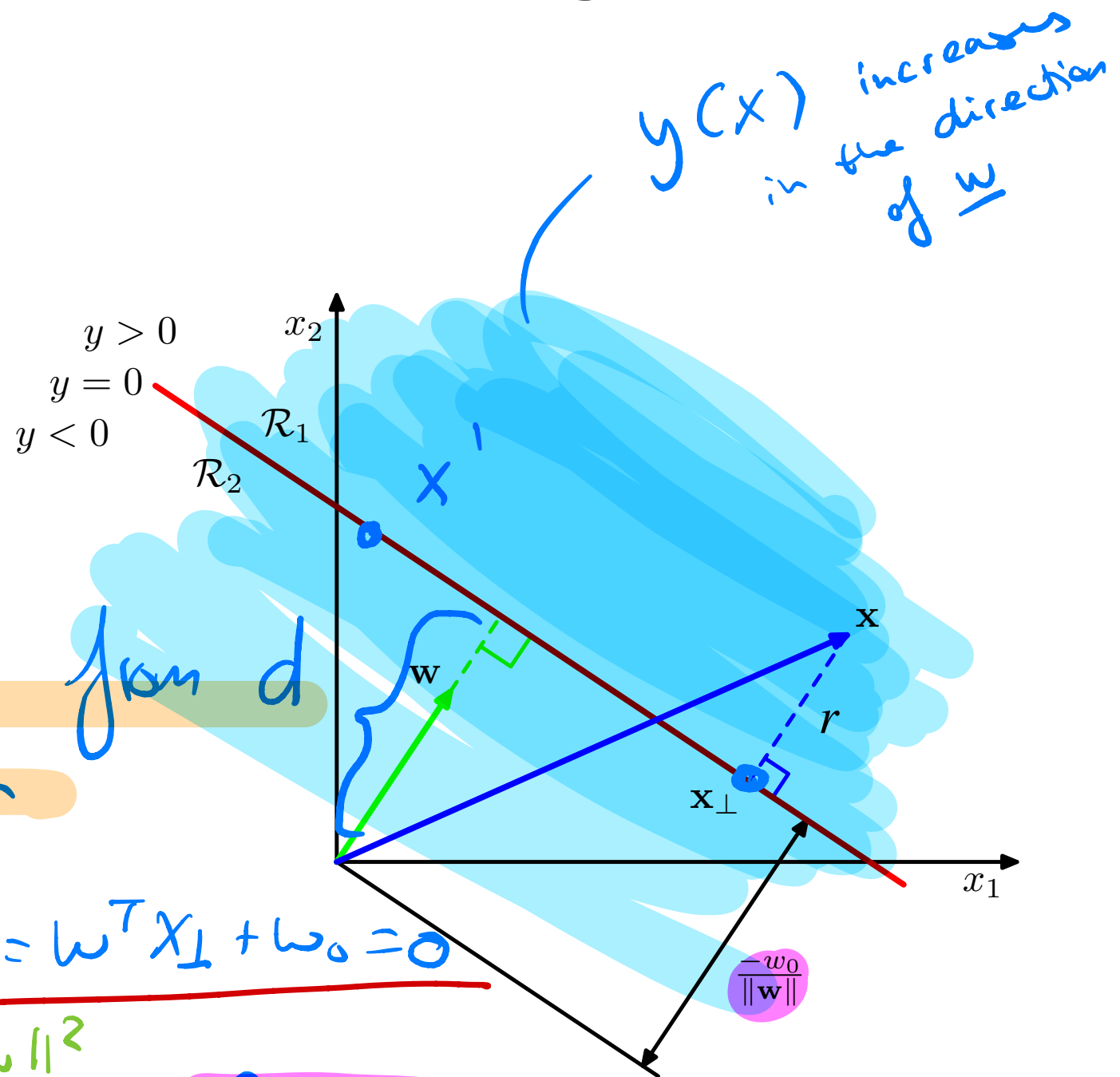
Then, $y(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0 = \mathbf{w}^T \mathbf{x}_\perp + r \frac{\mathbf{w}^T \mathbf{w}}{\|\mathbf{w}\|} + w_0 = r \cdot \|\mathbf{w}\|$

$y(\mathbf{x}_\perp) = \mathbf{w}^T \mathbf{x}_\perp + w_0 = 0$

$y(\mathbf{x}_\perp) = 0$



Distance r of a point to the d.b. is $r = \frac{y(\mathbf{x})}{\|\mathbf{w}\|}$



So the distance
of a point x to
the decision boundary
is proportional to $y(\underline{x})$

$$r = \frac{y(x)}{\|w\|}$$

Discriminant Functions: Multiple Classes

- ▶ K -class discriminant

$$y_k(\mathbf{x}) = \mathbf{w}_k^T \mathbf{x} + w_{k0}$$

like assigning to max prob

- ▶ Assign $\mathbf{x}$ to C_k if $\forall_{j \neq k} : y_k(\mathbf{x}) > y_j(\mathbf{x})$

- ▶ Decision boundary between R_k and R_j :

$$y_k(\mathbf{x}) = y_j(\mathbf{x}) \Leftrightarrow y_k(\mathbf{x}) - y_j(\mathbf{x}) = 0$$

linear d.b

$\Rightarrow$

$$(\underline{w}_k - \underline{w}_j) \underline{x} + (w_{k0} - w_{j0}) = 0$$

This linear

- ▶ Decision regions (for GLM) are convex

